GLP-1s and Reductions in Blood Pressure, HbA1c and Lipids in US Telehealth Patients

Shanmu Raja, MS*

November 12, 2026

Abstract

Background. Telehealth GLP-1 programs report outcomes from self-reported or program-ordered measurements, usually without a comparison group.
Methods. We linked 193,932 members of a U.S. telehealth obesity program to their outside clinical records through national health-information exchange (23.7 million readings). Members who started treatment later served as controls for those who started earlier. Members who started on branded and on compounded medication were analyzed as separate cohorts to avoid confounding; the study does not compare them.
Results. Six to twelve months after starting, systolic blood pressure was 4.9 mmHg lower than expected (branded cohort 5.1, compounded 4.9), HbA1c 0.21 points lower, LDL cholesterol 5 mg/dL lower, triglycerides 19 mg/dL lower and weight 22 lb lower. Regression to the mean was large: untreated members with high blood pressure fell 10 mmHg on their own, so most of the commonly reported improvement in that group is not a treatment effect. Placebo laboratory measures did not change, and program-recorded weights matched clinicians' weights. How much of the benefit comes from weight loss could not be determined.
Conclusions. Among members who stay on treatment, GLP-1 therapy through telehealth is followed by meaningful improvements in blood pressure, blood sugar and lipids, 11% to 28% smaller than simple before-and-after changes in the same data. If sustained, the blood pressure and LDL changes correspond in trial evidence to roughly 12% fewer major cardiovascular events. Programs should report outcomes measured outside the program, against a comparison group.

Keywords: GLP-1, semaglutide, tirzepatide, telehealth, health-information exchange, real-world evidence, regression to the mean, difference-in-differences.
JEL codes: I11, I12, I18.

1 Introduction

Telehealth companies now prescribe GLP-1 receptor agonists to hundreds of thousands of Americans, and employers, payers and regulators increasingly ask what those patients get for the money. The available evidence answers a narrower question than they ask. Digital and telehealth GLP-1 cohorts report weight, often self-reported, and sometimes laboratory values that the program itself ordered [Deanfield et al., 2026]. Real-world studies in health systems use clinician-recorded values but, with few exceptions, compare patients with themselves before and after treatment [Ruseva et al., 2025]. Before-after changes in risk factors that are selected for being high are dominated by regression

*Mochi Health. The author is employed by the telehealth program studied, which prescribes both branded and compounded GLP-1 medications; this is a conflict of interest. Outcomes are laboratory values and vital signs recorded by clinicians outside the program and retrieved through health-information exchange with member consent. Only aggregate tables leave the secure environment. Not peer reviewed.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

to the mean [Barnett et al., 2005], and the one large study that used patients not yet treated as controls found a systolic pressure effect of 1.1 mmHg, far below uncontrolled figures [Lu et al., 2025].

This paper makes three contributions. The first is the measurement design. Every outcome comes from a clinician outside the program, retrieved through national health-information exchange: 23.7 million readings across 168,819 members, with a median of 87 readings from two outside organizations among members with any reading. No GLP-1 or telehealth study we know of has measured outcomes this way, and the coverage figures are themselves evidence on how usable exchange data now are for outcomes research.

The second is a comparison group. Because members start treatment at different times and the exchange returns years of earlier records, members who start later can serve as controls for those who start earlier [Callaway and Sant'Anna, 2021, Borusyak et al., 2024, Batomen et al., 2024]. We apply this to every measure, check it against a simpler stacked difference-in-differences that uses identical selection rules for treated and control members, and test the design on placebo laboratory outcomes the drug should not move [Lipsitch et al., 2010].

The third is a cleaner design and a correction of claims. Members who started on branded medication and those who started on compounded medication differ in start dates, agent mix and potentially in payment and price, so we estimate every effect separately within each cohort rather than pooling them. The purpose is to keep formulation from confounding the estimates, not to compare the two; the paper makes no branded-versus-compounded comparison. We show how much of the commonly quoted out-of-range improvement is regression to the mean, and we show that the share of improvement attributable to weight loss, a question prominent since SELECT [Lincoff et al., 2023, Deanfield et al., 2025, Colhoun et al., 2026], cannot be pinned down with observational data of this kind.

2 Data

2.1 Linkage

Members of a U.S. telehealth obesity program operating in all 50 states consented to retrieval of their outside records through the Carequality and CommonWell networks. The program queries the networks for each consenting member and receives the structured results participating health systems hold: laboratory panels, vital signs and diagnoses. These readings come from care that would have happened without the program; no one at the program ordered, performed or recorded them. Linkage was run in mid to late 2026 for members active at that time: 212,886 members were linked, 193,932 of them with a dispensed GLP-1, and 168,819 had at least one outside reading. Among members with any outside reading, the median member has records from two outside organizations (75th percentile three) and 87 readings.

2.2 Cohorts

Figure 1 shows the study population. We classify members by the formulation of their first GLP-1 fill, using the dispensing pharmacy's compounding flag. 6,451 linked members started on a branded product (semaglutide or tirzepatide brands, or liraglutide), 2,820 of whom never received a compounded fill; 187,481 started on a compounded product. Table 1 describes both cohorts. Members were mostly women in their early forties with a body mass index of about 34 and near-normal average blood pressure, HbA1c and lipids; about a fifth had hypertension diagnosed before starting and about a sixth prediabetes.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

Table 1: Baseline characteristics of linked members, by cohort.

Characteristic	Branded	Compounded
Members	6,451	187,481
Age at first fill, mean (SD)	41.6 (12.9)	43.3 (12.5)
Female, %	81.3	84.4
Race or ethnicity reported, %	30.1	26.3
Non-Hispanic White, % of reported	63.9	67.4
Hispanic, % of reported	12.5	11.2
Black, % of reported	9.4	7.1
Asian, % of reported	4.8	3.8
Multiracial or other, % of reported	9.5	10.5
Tirzepatide at first fill, %	52.6	62.2
First fill in 2023 or earlier, %	5.8	1.6
First fill in 2024, %	19.6	13.9
First fill in 2025, %	20.0	33.1
First fill in 2026, %	54.7	51.4
Dispensed fills, median (IQR)	8 (3-19)	9 (5-16)
Outside organizations, median (IQR)	2 (1-4)	2 (1-3)
Outside readings, median (IQR)	52 (8-137)	59 (8-153)
BMI, kg/m^2, mean (SD) [n with baseline]	33.5 (7.4) [3,574]	34.2 (7.4) [105,504]
Weight, lb, mean (SD) [n with baseline]	204.6 (51.9) [3,681]	208.8 (51.2) [108,279]
Systolic BP, mmHg, mean (SD) [n with baseline]	123.0 (14.5) [3,718]	123.4 (14.3) [109,510]
Diastolic BP, mmHg, mean (SD) [n with baseline]	77.7 (9.7) [3,716]	77.8 (9.7) [109,496]
HbA1c, %, mean (SD) [n with baseline]	5.49 (0.68) [1,346]	5.44 (0.55) [42,868]
Fasting glucose, mg/dL, mean (SD) [n with baseline]	95.0 (17.8) [2,092]	95.1 (19.1) [64,058]
LDL-C, mg/dL, mean (SD) [n with baseline]	110.6 (32.4) [1,498]	112.2 (32.9) [49,751]
HDL-C, mg/dL, mean (SD) [n with baseline]	55.5 (15.4) [1,521]	55.6 (15.1) [50,050]
Triglycerides, mg/dL, mean (SD) [n with baseline]	120.2 (83.0) [1,520]	121.2 (72.4) [49,813]
Hypertension before start, %	18.8	21.0
Type 2 diabetes before start, %	3.4	3.1
Prediabetes before start, %	15.1	16.1
Dyslipidemia before start, %	24.7	26.6
Atherosclerotic cardiovascular disease before start, %	4.3	4.8
Fatty liver disease before start, %	3.3	3.8

Linked members with a dispensed GLP-1, by formulation of the first fill. Laboratory and vital-sign values are the last outside reading in the year before the first fill, among members with one; the number with a baseline value is in brackets. Diagnoses are outside diagnoses dated before the first fill. Race and ethnicity are self-reported to the program and missing for most members. The columns describe each cohort; they are not a comparison.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

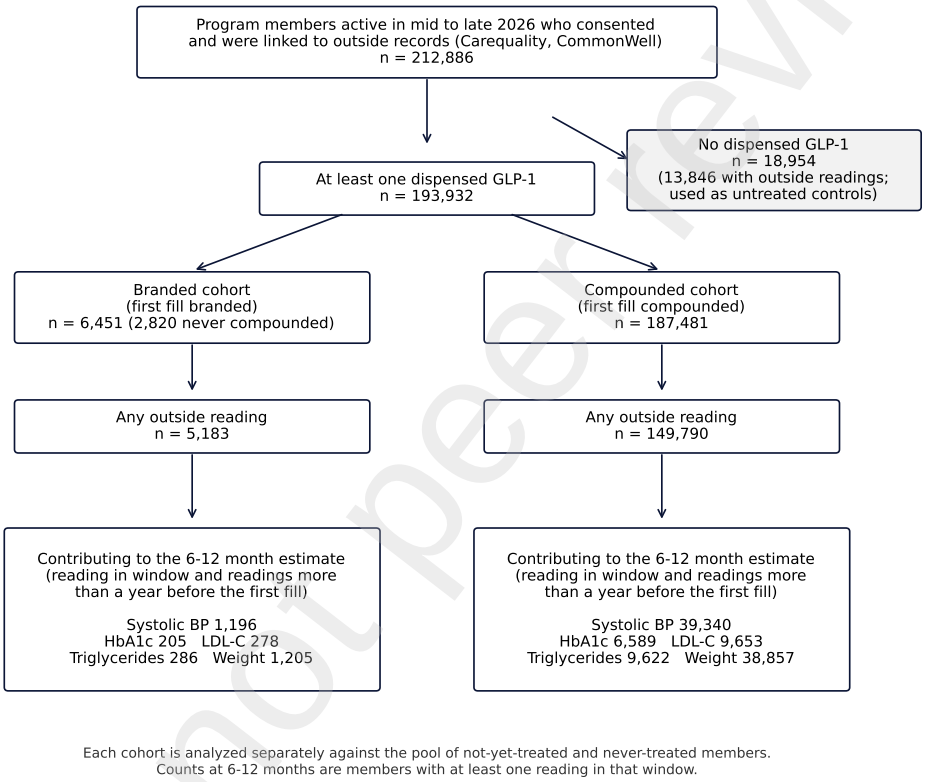


Figure 1: Study population. Counts at six to twelve months are members contributing at least one reading to that window of the imputation event study.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

2.3 Why the cohorts are analyzed separately

The two cohorts are not comparable populations. Branded starters received tirzepatide less often (53% against 62%) and their start dates are distributed differently across 2023 to 2026; the cohorts can also differ in how treatment is paid for, in price and in who chooses each option. Pooling them would let these differences, and any differences in what the medications are, mix into a single estimate. We therefore fit every model separately within each cohort, with each cohort's own treated members set against the full pool of not-yet-treated members, and report each cohort's estimates on their own. We do not estimate, test or interpret differences between the cohorts. The study is not designed to compare branded with compounded medication: the cohorts differ in ways the design does not balance, the branded cohort is small, and any contrast between them would be confounded. Cohorts appear in adjacent columns only to save space; the columns are not a comparison.

2.4 Measures

Readings were mapped from LOINC codes to systolic and diastolic pressure, HbA1c, fasting glucose, total, LDL and HDL cholesterol, triglycerides, ALT, eGFR, weight and body mass index, and to five placebo measures: sodium, potassium, hemoglobin, platelets and thyroid-stimulating hormone. Units were harmonized, values outside fixed plausibility ranges dropped (0.24% of readings), and same-day readings averaged. Diagnoses come from the same records; a member belongs to a diagnosed group if the diagnosis is dated before the first fill.

2.5 Selection on staying in the program

Because linkage covered members active in 2026, the linked cohort is selected on persistence. Among all members who started a GLP-1 by October 2025, 51% were still filling twelve months later (branded 48%, compounded 51%); among linked members the figure is 99%. Every estimate below describes members who stayed on treatment. They are comparable to per-protocol or persistent-user analyses, not to everyone who starts.

3 Methods

3.1 Imputation event study

For each measure we fit, on untreated readings only, a model with a member effect, a calendar-quarter effect and a five-year age-band effect:

$$y_{it} = \alpha_i + \lambda_{q(t)} + \kappa_{a(i,t)} + \varepsilon_{it}. \tag{1}$$

Untreated readings are those taken more than 365 days before a member's first fill. The member effect absorbs everything fixed about the member; the calendar effect absorbs secular trends and changes in testing, and is identified largely from members who start later; the age effect absorbs ageing beyond what the other two absorb. The fitted model imputes each later reading's untreated value, and the gap between observed and imputed is the effect for that reading [Borusyak et al., 2024]. We average gaps within member and time bin, then across members, so that frequently tested members do not dominate. Bins from 365 to 0 days before the first fill are not used in fitting and serve as placebo periods: they would show anticipation, enrollment-driven testing or a pre-existing trend.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

3.2 Stacked paired-change difference-in-differences

As a transparent check that speaks directly to regression to the mean, we stack the data by month of first fill. Treated members are those who started in that month; controls are linked members not yet on a GLP-1 at least a year later. Both groups get the same windows around the month: baseline is the last reading from 365 days before to 14 days after, follow-up the last reading 60 to 365 days after. We then apply the same selection rule to both groups (everyone with a pair, out of range at baseline, in range, or diagnosed before start) and report the treated change, the control change and their difference, weighted across months by treated counts, with a Poisson bootstrap over members. A selection rule that picks people at a high reading produces a fall in the control group too; the difference removes it. We also report the approach of Barnett et al. [2005]: select on the baseline reading but measure change from an earlier, independent reading taken 180 to 730 days before the start.

3.3 Placebo outcomes

Sodium and potassium are tightly regulated and should not change with GLP-1 treatment. Hemoglobin, platelets and TSH are less clean, because weight loss and reduced intake can move them, but none should show a step at initiation of the size seen in the target measures.

3.4 Weight share

To ask how much of each improvement runs through weight, we estimate the within-member slope of each measure on contemporaneous weight recorded by outside clinicians (same day for blood pressure, nearest weight within 30 days for labs), with member and calendar-quarter effects. We estimate it on untreated readings, so that the drug's direct effects cannot enter the slope, and separately on treated readings. Weight measurement error is corrected with a reliability ratio from pairs of weights recorded one to three days apart [Hutcheon et al., 2010]. The weight share is the slope times the imputed weight effect, divided by the imputed effect on the measure. We follow the AGReMA guidance in describing it as a decomposition under assumptions rather than a causal mediation estimate [Lee et al., 2021].

4 Results

4.1 Controlled effects

Figure 2 plots the imputed effects by time since first fill for each cohort in standardized units, and Table 2 reports the six-to-twelve-month bin. Systolic pressure falls by about 2.8 mmHg in the first quarter and by 4.9 mmHg at six to twelve months, and remains lower at one to two years (5.1 mmHg). HbA1c falls 0.21 points (0.25 in members diagnosed with diabetes or prediabetes), fasting glucose 7.1 mg/dL, LDL 5.1 mg/dL, total cholesterol 9.2 mg/dL and triglycerides 19.3 mg/dL. HDL dips during active weight loss and rises above its predicted level after the first year (1.8 mg/dL at one to two years), the pattern seen in weight-loss trials. Weight recorded by outside clinicians is 22.2 lb lower at six to twelve months and 28.9 lb lower at one to two years.

The placebo periods are close to zero. In the 90 days before the first fill, systolic pressure is 0.5 mmHg below its predicted value, a small drift; HbA1c is 0.06 points and triglycerides 2.6 mg/dL above, consistent with patients starting treatment when their values worsen, which if anything understates the effect measured from the pre-period. Adjusting for age changes the systolic estimate from 5.0 to 4.9 mmHg and matters only for weight itself (25.3 to 22.2 lb), as expected.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

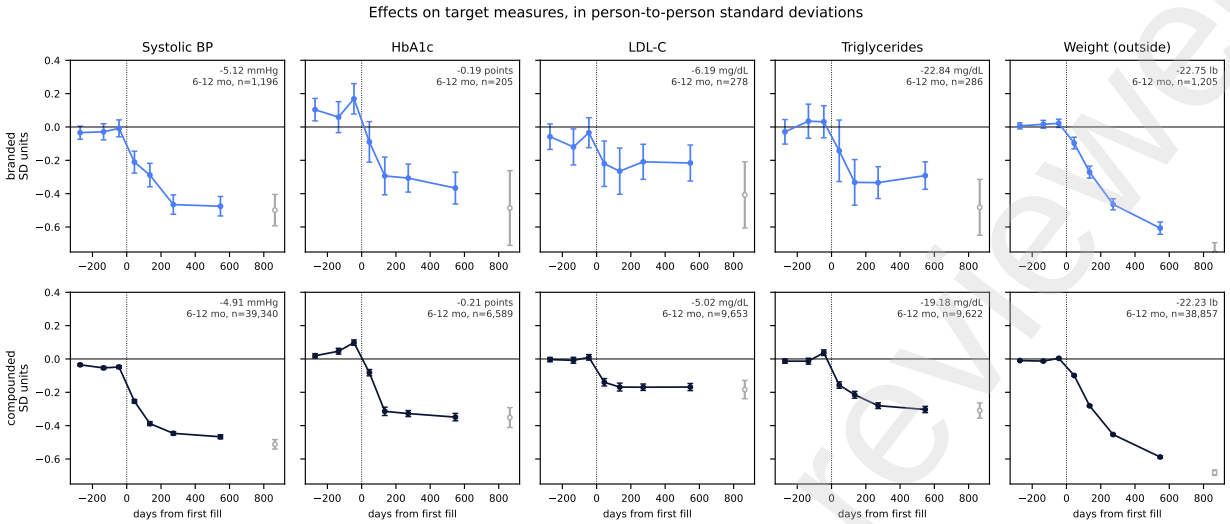


Figure 2: Effects on target measures by days from first GLP-1 fill, in units of each measure's person-to-person standard deviation (systolic BP 11.0 mmHg, HbA1c 0.63 points, LDL-C 29.6 mg/dL, triglycerides 68.4 mg/dL, weight 49.0 lb). Each cohort is estimated separately and shown in its own row; the rows are not a comparison. Bins before day 0 are placebo periods. The last bin (730 to 1,000 days), which few members have reached, is shown hollow and grey. Bars are 95% intervals; labels give the six-to-twelve-month effect in natural units.

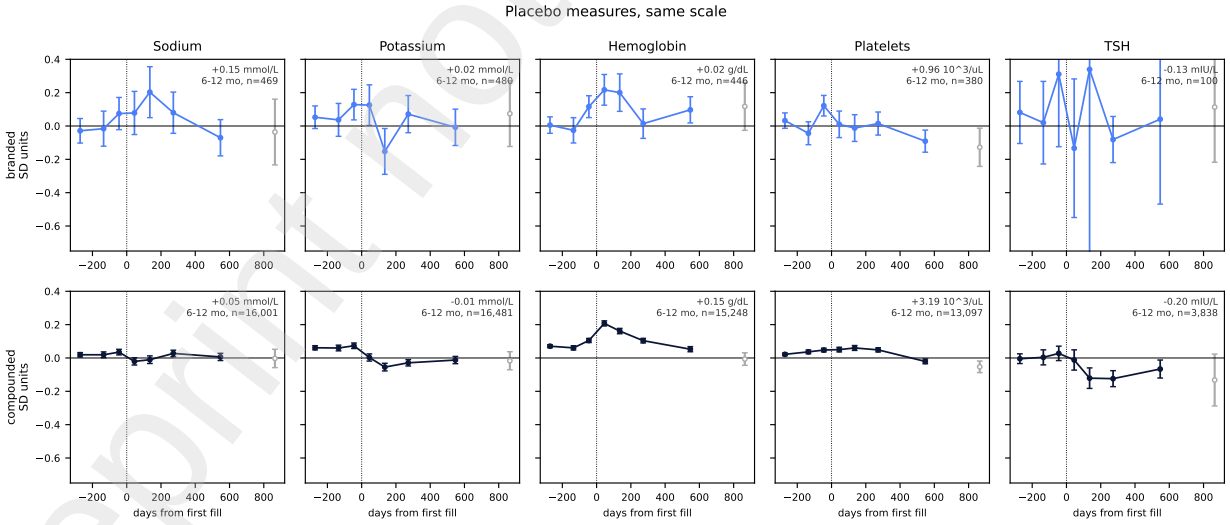


Figure 3: Placebo measures on the same standardized scale as Figure 2 (person-to-person standard deviations: sodium 1.83 mmol/L, potassium 0.32 mmol/L, hemoglobin 1.44 g/dL, platelets 65.6 $\times 10^3/\mu$L, TSH 1.58 mIU/L).

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

Table 2: Controlled effects six to twelve months after the first GLP-1 fill (imputation event study).

Measure	All linked	Branded	Branded only	Compounded
Systolic BP (mmHg)	−4.92 (0.06)	−5.12 (0.33)	−5.92 (0.81)	−4.91 (0.06)
Diastolic BP (mmHg)	−2.28 (0.04)	−2.57 (0.23)	−2.01 (0.57)	−2.27 (0.04)
HbA1c (points)	−0.21 (0.01)	−0.19 (0.03)	−0.23 (0.04)	−0.21 (0.01)
diagnosed diabetes or prediabetes	−0.25 (0.01)	−0.22 (0.05)	−0.23 (0.06)	−0.25 (0.01)
Fasting glucose (mg/dL)	−7.08 (0.13)	−7.68 (0.83)	−6.57 (1.63)	−7.06 (0.13)
LDL-C (mg/dL)	−5.05 (0.29)	−6.19 (1.59)	−8.52 (3.66)	−5.02 (0.29)
Triglycerides (mg/dL)	−19.3 (0.6)	−22.8 (3.3)	−34.4 (8.9)	−19.2 (0.6)
Weight, outside (lb)	−22.2 (0.1)	−22.7 (0.8)	−24.5 (2.0)	−22.2 (0.1)
Placebo outcomes, effect at 6–12 months / placebo period 90–0 days before				
Sodium (mmol/L)	0.05 / 0.07	0.15 / 0.14		0.05 / 0.06
Potassium (mmol/L)	−0.01 / 0.02	0.02 / 0.04		−0.01 / 0.02
Platelets ($10^3/\mu$L)	3.1 / 3.3	1.0 / 8.0		3.2 / 3.1

Standard errors in parentheses. Members contributing at six to twelve months: systolic pressure 40,536 (branded 1,196, branded only 215); HbA1c 6,794 (205, 42); LDL 9,931 (278, 59). Branded: first fill branded; branded only: no compounded fill ever. Placebo outcomes should show no step at the first fill: the comparison is the effect at six to twelve months against the same model's value in the 90 days before the fill.

4.2 Regression to the mean

Figure 4 shows the stacked difference-in-differences for systolic pressure across the range of baseline values, and Appendix Table B1 gives the results for five measures in each cohort; the pooled figures below combine both cohorts. Among members out of range at baseline, systolic pressure fell 13.9 mmHg in those who started a GLP-1, the size of figure commonly quoted for such programs. In members selected the same way who had not started, it fell 10.1 mmHg. Among members in range at baseline, it rose 3.6 mmHg in controls. The difference is 3.7 mmHg in every selection: out of range, in range, everyone, and 3.3 mmHg among members diagnosed with hypertension. The same holds for LDL (out-of-range treated −21.4, controls −16.3, difference −5.1; everyone −4.8) and fasting glucose (out-of-range difference −7.5, everyone −5.5). Measuring change from an earlier independent reading, as Barnett et al. [2005] recommend, halves the out-of-range systolic fall from 13.9 to 6.8 mmHg. Across everyone with a pair, the controlled effects are smaller than the treated members' raw before-after changes by 14% for systolic pressure (−3.7 against −4.3 mmHg), 14% for HbA1c, 27% for LDL, 21% for triglycerides and 11% for glucose, and by 28% for systolic pressure among members diagnosed with hypertension. The same pattern appears within each cohort analyzed separately: in the branded cohort the controlled systolic effect is −4.5 mmHg for everyone with a pair and −4.6 in the out-of-range subgroup, against a 10.1 mmHg fall in controls; in the compounded cohort it is −3.7 and −3.7. Branded estimates for triglycerides and glucose rest on a few hundred members and are imprecise. No branded member with a baseline HbA1c of 6.5% or more had a pair, so the branded out-of-range HbA1c estimate is missing.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

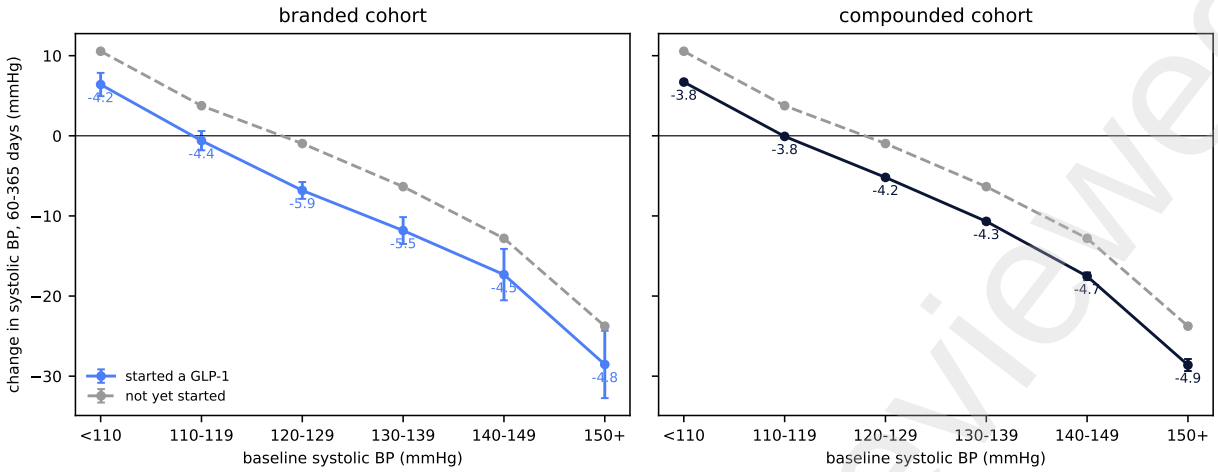


Figure 4: Change in systolic blood pressure from baseline to 60–365 days, by baseline systolic pressure, in members who started a GLP-1 and in not-yet-treated members given the same windows, each cohort estimated separately. Numbers are the treated-minus-control difference in each band. Untreated members with high baseline readings fall, and those with low readings rise, by about as much as treated members; the difference stays near 4 mmHg across the range. Bars are 95% intervals.

4.3 Placebo outcomes

On the same standardized scale (Figure 3), the placebo measures stay within about 0.2 standard deviations of zero, apart from imprecise branded TSH estimates resting on about 100 members, against falls of 0.2 to 0.6 standard deviations in the target measures. Sodium and potassium show no step at initiation (Table 2): sodium is 0.05 mmol/L above its predicted value at six to twelve months against 0.07 in the 90 days before, and potassium moves by less than 0.03 mmol/L throughout. Platelets are as far above prediction before the fill as after, and decline only after the first year, in line with the known association of platelet counts with adiposity. Hemoglobin rises by about 0.3 g/dL in the first three months and returns toward its prediction (0.15 g/dL at six to twelve months), consistent with reduced fluid and food intake early in treatment, and TSH falls slightly, consistent with weight loss. None shows a step comparable to the target measures, and the two strictly regulated placebos do not move.

4.4 Program weights and outside weights

Program-recorded weights agree with weights recorded by outside clinicians within seven days (Figure 5; 95,694 pairs from 49,278 members): the mean difference is 0.9 lb, the correlation 0.96, and 69% of pairs fall within 5 lb. Weight change agrees too: among 15,199 members with two such pairs at least 120 days apart, program weights fell 20.5 lb and outside weights 19.8 lb. In the linked cohort, program weights fell 10.2% at six months and 14.3% at twelve, with 47% losing at least 15%; the branded and compounded cohorts are within 0.1 points of each other.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

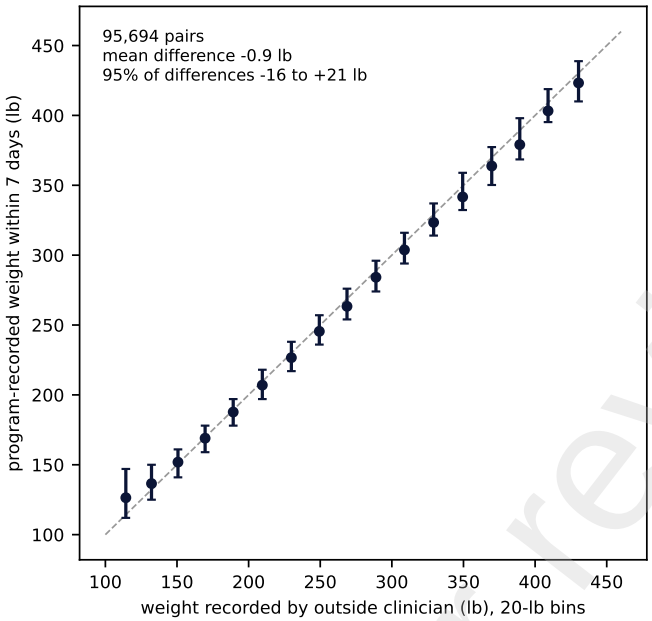


Figure 5: Program-recorded weight against weight recorded by an outside clinician within seven days, averaged within 20-lb bins of outside weight (bins with at least 30 pairs). Bars span the 10th to 90th percentile of program weights in each bin; the dashed line is equality.

4.5 By agent

Appendix Table B2 splits the six-to-twelve-month effects by agent within each cohort. In both cohorts tirzepatide shows larger effects than semaglutide for weight, systolic pressure, HbA1c and triglycerides, and similar or smaller effects for LDL. The ordering matches the trials and holds within the branded cohort as well as the compounded one, a further check that the estimates track pharmacology rather than enrollment.

4.6 Effects by sex, age, body mass index, race and ethnicity

Figure 6 and Appendix Table A1 report the six-to-twelve-month effects by sex, age at first fill, baseline body mass index and self-reported race and ethnicity, within each cohort. In both cohorts, effects are present in every group with enough members to estimate them. Systolic effects are larger in men and grow with age in the compounded cohort, from 4.1 mmHg at 18 to 34 to 6.0 mmHg at 65 and over, and HbA1c effects grow with age from 0.12 to 0.31 points, in line with higher baseline values in older members. LDL and triglyceride effects are larger in men (in the compounded cohort, LDL −14.1 against −3.4 mg/dL); because we do not observe outside prescriptions, part of this difference could reflect lipid-lowering treatment started around the same time rather than the GLP-1. Effects on blood pressure and HbA1c are similar across body mass index groups; lipid effects are smaller at a body mass index of 40 or more. The branded cohort's subgroups are small (156 men; 46 members aged 65 or over for blood pressure) and their estimates imprecise. By race and ethnicity, systolic and HbA1c effects are present in every group; in the compounded cohort the systolic effect is 5.3 mmHg in non-Hispanic White members, 4.1 in Hispanic, 3.5 in Black and 4.7 in Asian members, and LDL and triglyceride effects are small in Black members (LDL +0.7, triglycerides −4.6 mg/dL). Race and ethnicity are self-reported for only about a quarter of members,

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

and those who report may differ from those who do not, so these estimates describe the reporting members only. Weight effects by baseline body mass index are not reported, because selecting members on a baseline weight-derived value produces regression to the mean in weight itself.

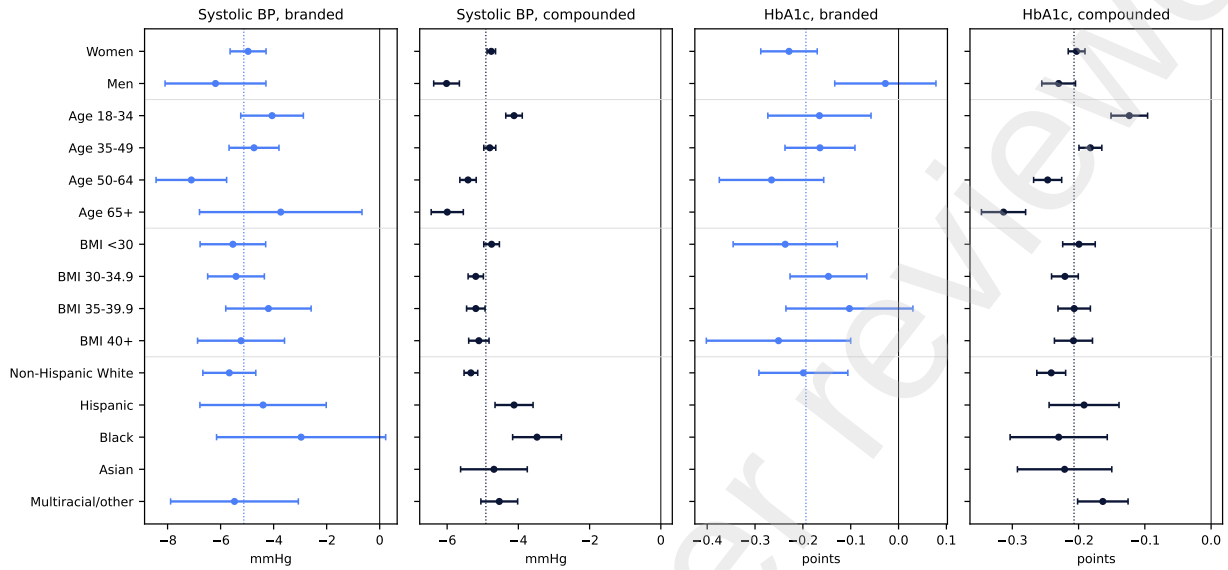


Figure 6: Effects on systolic blood pressure and HbA1c at six to twelve months by subgroup, each cohort estimated separately. Dotted lines mark the cohort's overall effect. Groups with fewer than 30 members are omitted. Bars are 95% intervals; full results, including LDL-C, triglycerides and weight, are in Appendix Table A1.

4.7 How much runs through weight

Table 3 reports the weight decomposition by cohort. Measured with outside clinicians' scales and estimated on untreated readings, a pound of weight moves systolic pressure by about 0.08 to 0.09 mmHg within member; applied to the 22-pound weight effect, this accounts for 40% of the systolic effect in the branded cohort and 37% in the compounded cohort, each estimated separately (45% and 42% with the slope estimated on treated readings). Pooled, the shares are 36% and 42%. The shares are higher for HbA1c and triglycerides and lower for glucose, within each cohort. Correcting for weight measurement error changes the shares by one to five points, because the reliability of outside weights is high (0.89 to 0.97).

These numbers should not be read as the answer. The within-member slope measures how pressure moves with ordinary fluctuations in weight, about 0.18 mmHg per kilogram. The randomized-trial literature on sustained weight reduction finds about 1 mmHg per kilogram [Neter et al., 2003], which applied to our 10-kilogram weight effect would account for more than the whole systolic effect, and the mediation analysis of SURMOUNT-1 attributes 68% of tirzepatide's placebo-adjusted systolic effect to weight loss [Krumholz et al., 2024]. The weight share of systolic pressure in this population could therefore be anywhere from about a third to all of it, depending on whether short-run weight fluctuations and sustained loss act alike, which these data cannot test. In SELECT the share of cardiovascular benefit mediated by body weight was estimated at about a fifth, with a confidence interval wide enough to include none or all of it [Colhoun et al., 2026]. We conclude that real-world data, even with third-party weights, do not identify the weight-mediated share, and that

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

single-number shares in observational studies should be treated with caution.

Table 3: Share of each six-to-twelve-month effect attributable to weight, by cohort and by slope used.

	Branded		Compounded	
Measure	untreated slope	treated slope	untreated slope	treated slope
Systolic BP	40%	45%	37%	42%
Diastolic BP	48%	36%	49%	45%
HbA1c	91%	69%	74%	60%
Fasting glucose	34%	51%	32%	23%
Triglycerides	78%	58%	90%	70%
LDL-C	60%	-9%	72%	34%

Slopes are within-member, with member and quarter effects, on outside-recorded weight, corrected for measurement error with reliability ratios from repeat weights 1 to 3 days apart (0.97 untreated, 0.89 treated). Untreated: readings more than 90 days before the first fill; treated: 60 or more days after. Branded slopes rest on 205 to 4,209 members and are imprecise; the negative branded LDL share from the treated slope reflects a slope near zero. For comparison, Neter et al. [2003] imply about 0.45 mmHg systolic per lb of sustained loss, more than five times the untreated slope.

5 Discussion

Three findings matter for how telehealth GLP-1 outcomes should be read.

First, the improvements hold up within each cohort analyzed. With a not-yet-treated comparison, a placebo check and outside measurement, members who stay on treatment show a systolic effect of about 4 to 5 mmHg, an HbA1c effect of about a quarter point in those with dysglycemia, and LDL and triglyceride effects of 5 and 19 mg/dL. The branded cohort, analyzed on its own, shows effects of the same order (systolic −5.1 mmHg, HbA1c −0.19, LDL −6.2 mg/dL), with wider intervals; these are not a comparison with the compounded cohort, which differs in timing, agent mix and potentially payment. Our systolic effect lies between the 1.1 mmHg found with a similar not-yet-treated design in 23,522 health-system patients prescribed semaglutide, who lost 3.8% of body weight at 13 to 24 months and were counted whether or not they filled or persisted [Lu et al., 2025], and the 6.8 mmHg placebo-adjusted effect of tirzepatide in SURMOUNT-1, where weight fell 15% to 21% [Krumholz et al., 2024]. Our members persisted and lost about 14% of body weight, and the ordering of the three estimates follows the ordering of weight loss.

Second, the figures most often quoted for programs like this are mostly regression to the mean. The out-of-range subgroup change in systolic pressure is 13.9 mmHg in treated members and 10.1 mmHg in untreated members selected the same way. Reports that select on a high baseline reading and present the fall as an outcome overstate the effect roughly fourfold. The controlled effect is the same, about 3.7 mmHg, whichever way the group is chosen.

Third, the weight share is not identified. The answer depends on which slope of pressure on weight one believes, and the two most defensible choices differ by a factor of five. Observational estimates that report most of the benefit as arriving "beyond weight" should be read in that light.

5.1 What changes of this size mean for health

Changes of this size matter for health if they are sustained. The largest pooled analysis of blood-pressure-lowering trials, 344,716 participants in 48 randomized trials, found that each 5 mmHg

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

lower systolic pressure cut major cardiovascular events (stroke, myocardial infarction, ischemic heart disease and heart failure) by about 10%, with hazard ratios of 0.91 without and 0.89 with prior cardiovascular disease, and with no reliable difference across baseline pressures, including normal and high-normal ones [Blood Pressure Lowering Treatment Trialists' Collaboration, 2021]. Applied to the 4.9 mmHg effect here, that relationship corresponds to about 9% fewer major cardiovascular events, and to about 10% among members diagnosed with hypertension (5.7 mmHg). For LDL cholesterol, each 1.0 mmol/L (38.7 mg/dL) reduction cut major vascular events by about 22% [Cholesterol Treatment Trialists' Collaboration, 2010]; the 5 mg/dL effect here corresponds to about 3%. Taken together and assuming the two act independently, the blood pressure and LDL effects correspond to roughly 12% fewer major cardiovascular events.

Two features make this relevant at population scale. First, the effects appear in members whose average baseline values were close to normal (systolic pressure 123 mmHg, LDL 111 mg/dL), the range in which the trial evidence shows the same proportional benefit but where few people are treated for blood pressure or cholesterol. Second, they arise from GLP-1 treatment prescribed through telehealth in ordinary use, outside a trial and without a trial's structured follow-up. Because relative risk reductions of this size apply to everyone treated, small shifts across a large population prevent more events than large shifts in a small high-risk group.

These figures are projections from trials, not events counted in this cohort, and they rest on three assumptions the data do not test: that the changes persist beyond the one to two years observed, that a blood-pressure or LDL reduction produced alongside a GLP-1 confers the same benefit as one produced by antihypertensive or statin therapy, and that members' other treatment did not change. The estimated effect is the effect of starting treatment through the program, which includes the medication, the program's care, and any concurrent changes in members' outside care that the records do not capture; it is not an estimate of the medication alone.

5.2 Limitations

The largest limitation is selection on persistence: records were linked for members active in 2026, so the estimates describe people who stayed on treatment and cannot speak to those who stopped, or test whether improvements reverse when treatment ends [Horn et al., 2026]. Only half of all members are still filling at twelve months. Outside readings are taken when patients see clinicians, so testing frequency is informative [Goldstein et al., 2016]; the member and calendar effects and the placebo outcomes address this partly, not fully. Controls are members who later start a GLP-1, which helps comparability but means the calendar effects are estimated from future starters. The branded cohort is small, so its estimates, and especially its later bins, are imprecise. The study does not compare branded and compounded medication; a comparison would need a design that balances payment, price, timing and agent, which this one does not attempt. Diagnosis-based groups depend on what outside systems record. We do not observe outside prescriptions consistently, so changes in antihypertensive or statin use are not controlled. The program's own commercial interest in the result is disclosed in the declarations.

6 Conclusion and recommendations

Measured by patients' own outside clinicians and compared with patients who had not yet started, GLP-1 treatment delivered through telehealth is followed by clinically meaningful improvements in blood pressure, glycemia and lipids among those who stay on it, in each of the two cohorts analyzed separately, and 11% to 28% smaller than the same data's uncontrolled before-after changes. We recommend that telehealth programs, payers and employers (1) require outcomes measured

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

outside the program and a comparison with not-yet-treated patients, (2) stop reporting changes in subgroups selected for high baseline values, which are dominated by regression to the mean, (3) report persistence alongside outcomes, because outcome estimates among persisters do not describe everyone who starts, and (4) treat observational claims about how much benefit arrives through weight as unidentified. The next study should link records at enrollment rather than for current members, so that outcomes after stopping treatment can be measured.

Declarations

Ethics. This is a retrospective analysis of existing program and outside clinical records, carried out within the program's data governance; it was not reviewed by an institutional review board. Members consented to retrieval of their outside records through health-information exchange.

Data protection. Analyses were run inside the program's secure environment on identifiable records; only aggregate results leave it, and results resting on fewer than 20 members are not reported. No patient-level data are shared.

Data availability. The aggregate tables underlying every result (listed in the data appendix) are available from the author. Patient-level data are protected health information; requests for access can be made to the author and are subject to a data use agreement and applicable privacy law.

Code availability. Analysis code is available from the author on request.

Funding. No external funding. The work was carried out by an employee of Mochi Health during employment.

Competing interests. The author is employed by Mochi Health, the telehealth program studied, which prescribes both branded and compounded GLP-1 medications and has a commercial interest in reported outcomes. Mochi Health approved release of the aggregate data.

Author contributions. S.R. designed the study, carried out the analyses and wrote the manuscript.

Reporting guideline. The study is reported following STROBE for cohort studies; the completed checklist is available from the author.

Appendix A: Effects by subgroup

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

Table A1: Effects at six to twelve months by subgroup, branded cohort (top) and compounded cohort (bottom), each estimated separately.

Group	Systolic BP	HbA1c	LDL-C	Triglycerides	Weight
Branded cohort					
Women	−5.0 (0.3) [1,040]	−0.23 (0.03) [169]	−5.7 (1.6) [234]	−21.3 (3.4) [241]	−23.1 (0.9) [1,051]
Men	−6.2 (1.0) [156]	−0.03 (0.05) [36]	−8.8 (5.2) [44]	−31.0 (11.4) [45]	−20.5 (2.4) [154]
Age 18–34	−4.1 (0.6) [245]	−0.17 (0.06) [35]	−3.0 (3.7) [35]	−18.9 (9.1) [36]	−20.4 (2.0) [242]
Age 35–49	−4.7 (0.5) [597]	−0.16 (0.04) [101]	−6.3 (2.1) [134]	−24.0 (4.9) [138]	−24.0 (1.2) [602]
Age 50–64	−7.1 (0.7) [298]	−0.27 (0.06) [62]	−7.1 (3.2) [95]	−23.0 (5.9) [97]	−23.1 (1.6) [305]
Age 65+	−3.7 (1.6) [46]	–	–	–	−19.6 (2.4) [46]
BMI <30	−5.5 (0.6) [258]	−0.24 (0.06) [37]	−6.3 (3.4) [73]	−26.8 (4.7) [73]	–
BMI 30–34.9	−5.4 (0.5) [331]	−0.15 (0.04) [60]	−8.1 (3.2) [83]	−21.9 (6.9) [88]	–
BMI 35–39.9	−4.2 (0.8) [215]	−0.10 (0.07) [41]	−4.7 (3.4) [52]	−19.6 (9.6) [51]	–
BMI 40+	−5.2 (0.8) [186]	−0.25 (0.08) [44]	−2.7 (2.6) [48]	−14.0 (6.9) [49]	–
Non-Hispanic White	−5.7 (0.5) [475]	−0.20 (0.05) [82]	−7.7 (2.3) [129]	−19.7 (4.7) [132]	−26.0 (1.4) [477]
Hispanic	−4.4 (1.2) [67]	–	–	–	−21.2 (2.7) [64]
Black	−3.0 (1.6) [74]	–	–	–	−27.1 (3.6) [75]
Asian	–	–	–	–	–
Multiracial or other	−5.5 (1.2) [64]	–	–	–	−27.7 (3.1) [70]
Race not reported	−5.0 (0.5) [490]	−0.17 (0.04) [84]	−6.7 (2.5) [103]	−25.4 (6.5) [103]	−18.7 (1.2) [493]
Compounded cohort					
Women	−4.8 (0.1) [34,571]	−0.20 (0.01) [5,646]	−3.4 (0.3) [8,215]	−16.3 (0.6) [8,169]	−22.8 (0.1) [34,186]
Men	−6.0 (0.2) [4,769]	−0.23 (0.01) [943]	−14.1 (0.8) [1,438]	−35.4 (2.1) [1,453]	−17.9 (0.4) [4,671]
Age 18–34	−4.1 (0.1) [8,176]	−0.12 (0.01) [1,030]	−1.6 (0.7) [1,242]	−20.1 (1.5) [1,268]	−21.7 (0.3) [7,840]
Age 35–49	−4.8 (0.1) [17,072]	−0.18 (0.01) [2,733]	−5.4 (0.4) [3,993]	−20.5 (1.0) [3,939]	−23.3 (0.2) [16,821]
Age 50–64	−5.4 (0.1) [10,894]	−0.25 (0.01) [2,142]	−5.5 (0.6) [3,362]	−18.4 (1.0) [3,352]	−21.5 (0.2) [10,957]
Age 65+	−6.0 (0.2) [3,048]	−0.31 (0.02) [667]	−6.2 (0.9) [1,046]	−15.6 (1.5) [1,054]	−21.4 (0.4) [3,081]
BMI <30	−4.8 (0.1) [9,293]	−0.20 (0.01) [1,436]	−6.2 (0.6) [2,429]	−19.7 (1.1) [2,418]	–
BMI 30–34.9	−5.2 (0.1) [10,075]	−0.22 (0.01) [1,826]	−5.7 (0.6) [2,759]	−20.2 (1.2) [2,753]	–
BMI 35–39.9	−5.2 (0.1) [7,222]	−0.21 (0.01) [1,323]	−5.0 (0.7) [1,879]	−21.7 (1.4) [1,867]	–
BMI 40+	−5.1 (0.1) [6,589]	−0.21 (0.01) [1,380]	−3.2 (0.6) [1,775]	−15.2 (1.4) [1,767]	–
Non-Hispanic White	−5.3 (0.1) [13,116]	−0.24 (0.01) [2,033]	−5.7 (0.5) [3,361]	−21.4 (0.9) [3,364]	−25.3 (0.2) [13,012]
Hispanic	−4.1 (0.3) [1,653]	−0.19 (0.03) [293]	−2.7 (1.3) [382]	−19.9 (3.3) [365]	−20.6 (0.6) [1,617]
Black	−3.5 (0.3) [1,208]	−0.23 (0.04) [251]	0.7 (1.5) [298]	−4.6 (2.0) [301]	−22.4 (0.8) [1,203]
Asian	−4.7 (0.5) [550]	−0.22 (0.04) [131]	−3.5 (2.3) [156]	−35.0 (6.5) [151]	−16.4 (0.9) [522]
Multiracial or other	−4.5 (0.3) [1,854]	−0.16 (0.02) [292]	−9.6 (1.4) [430]	−20.3 (2.7) [427]	−24.0 (0.6) [1,813]
Race not reported	−4.8 (0.1) [20,959]	−0.19 (0.01) [3,589]	−4.7 (0.4) [5,026]	−18.0 (0.9) [5,014]	−20.4 (0.2) [20,690]

Effect (standard error) [members]. Systolic BP in mmHg, HbA1c in points, LDL-C and triglycerides in mg/dL, weight (outside clinicians) in lb. Cells with fewer than 30 members are blank. Weight by baseline body mass index is not shown, because selection on a weight-derived value induces regression to the mean in weight. Age at first fill; body mass index from the last outside reading in the year before the first fill; race and ethnicity self-reported to the program.

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

Appendix B: Additional results

Table B1: Paired changes, treated / not-yet-treated controls / **difference**, 60 to 365 days, by cohort.

Measure	Selection	Branded	Compounded
Systolic BP (mmHg)	everyone with a pair	−5.12 / −0.63 / **−4.49** (1,488)	−4.31 / −0.62 / **−3.69** (53,902)
	out of range at baseline	−14.68 / −10.07 / **−4.61** (453)	−13.83 / −10.13 / **−3.70** (16,649)
	diagnosed before start	−5.89 / −1.17 / **−4.72** (436)	−4.50 / −1.27 / **−3.23** (18,150)
HbA1c (points)	everyone with a pair	−0.27 / −0.04 / **−0.23** (266)	−0.29 / −0.04 / **−0.25** (10,508)
	out of range at baseline	–	−1.17 / −0.30 / **−0.86** (356)
	diagnosed before start	−0.30 / −0.04 / **−0.26** (93)	−0.36 / −0.05 / **−0.31** (5,432)
LDL-C (mg/dL)	everyone with a pair	−6.47 / −1.98 / **−4.49** (340)	−6.55 / −1.77 / **−4.78** (13,977)
	out of range at baseline	−23.93 / −16.70 / **−7.24** (40)	−21.42 / −16.32 / **−5.10** (4,152)
	diagnosed before start	−7.95 / −3.22 / **−4.73** (150)	−7.82 / −2.91 / **−4.91** (8,484)
Triglycerides (mg/dL)	everyone with a pair	−16.92 / −4.84 / **−12.08** (344)	−21.94 / −4.53 / **−17.41** (13,993)
	out of range at baseline	–	−71.36 / −39.66 / **−31.70** (3,651)
	diagnosed before start	−12.71 / −6.36 / **−6.35** (156)	−25.04 / −6.19 / **−18.86** (8,512)
Fasting glucose (mg/dL)	everyone with a pair	−5.45 / −0.89 / **−4.56** (532)	−6.18 / −0.70 / **−5.47** (21,143)
	out of range at baseline	−18.38 / −12.92 / **−5.46** (104)	−20.30 / −12.79 / **−7.51** (5,955)
	diagnosed before start	−7.12 / −0.88 / **−6.24** (169)	−8.16 / −0.53 / **−7.64** (7,625)

Treated members in parentheses. Controls are linked members not yet on a GLP-1 at least a year after each treated month, given the same windows and the same selection rule. Out-of-range thresholds: systolic 130 mmHg, HbA1c 6.5%, LDL 130 mg/dL, triglycerides 150 mg/dL, glucose 100 mg/dL. Cells with fewer than 20 treated members are left blank (the branded HbA1c and triglyceride out-of-range cells). Bootstrap standard errors for systolic pressure are 0.09 to 0.19 mmHg in the compounded cohort and 0.39 to 0.88 mmHg in the branded cohort; all are available from the author.

Table B2: Effects at six to twelve months by agent and cohort (imputation event study).

	Branded		Compounded	
Measure	Tirzepatide	Semaglutide	Tirzepatide	Semaglutide
Weight, outside (lb)	−25.2 (1.1)	−17.8 (1.2)	−25.5 (0.2)	−17.9 (0.2)
Systolic BP (mmHg)	−5.55 (0.41)	−4.34 (0.53)	−5.53 (0.08)	−4.11 (0.09)
HbA1c (points)	−0.21 (0.03)	−0.14 (0.04)	−0.24 (0.01)	−0.16 (0.01)
Triglycerides (mg/dL)	−24.4 (4.2)	−19.7 (5.6)	−20.6 (0.8)	−17.2 (0.9)
LDL-C (mg/dL)	−5.8 (1.9)	−7.6 (2.8)	−4.7 (0.4)	−5.5 (0.4)
Members (systolic BP)	791	399	22,165	17,175

Agent at first fill. Standard errors in parentheses.

Data appendix

The following aggregate tables are available from the author: the imputation event study for all measures, cohorts and samples (`imputation_event_study.csv`), the stacked difference-in-differences and Barnett estimates for systolic pressure, HbA1c, LDL, triglycerides and glucose (`rtm_stacked_did.csv`), the weight decomposition (`mediation_outside_weight.csv`), program outcomes and cohort descriptors (`telehealth_outcomes.csv`), persistence for all members (`persistence_all_members.csv`), weight agreement (`weight_agreement.csv`), record coverage

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

(`coverage.csv`), agent comparisons (`specificity.csv`, `agent_by_cohort.csv`), baseline characteristics (`baseline_characteristics.csv`), subgroup effects (`subgroups.csv`) and the standard deviations used in the figures (`metric_sd.csv`). Patient-level files remain in the program's secure environment.

References

Adrian G. Barnett, Jolieke C. van der Pols, and Annette J. Dobson. Regression to the mean: What it is and how to deal with it. *International Journal of Epidemiology*, 34(1):215–220, 2005. doi: 10.1093/ije/dyh299.

Brice Batomen et al. Staggered interventions with no control groups. *International Journal of Epidemiology*, 53(6):dyae137, 2024. doi: 10.1093/ije/dyae137.

Blood Pressure Lowering Treatment Trialists' Collaboration. Pharmacological blood pressure lowering for primary and secondary prevention of cardiovascular disease across different levels of blood pressure: An individual participant-level data meta-analysis. *The Lancet*, 397(10285): 1625–1636, 2021. doi: 10.1016/S0140-6736(21)00590-0.

Kirill Borusyak, Xavier Jaravel, and Jann Spiess. Revisiting event-study designs: Robust and efficient estimation. *Review of Economic Studies*, 91(6):3253–3285, 2024. doi: 10.1093/restud/rdae007.

Brantly Callaway and Pedro H. C. Sant'Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021. doi: 10.1016/j.jeconom.2020.12.001.

Cholesterol Treatment Trialists' Collaboration. Efficacy and safety of more intensive lowering of ldl cholesterol: A meta-analysis of data from 170,000 participants in 26 randomised trials. *The Lancet*, 376(9753):1670–1681, 2010. doi: 10.1016/S0140-6736(10)61350-5.

Helen M. Colhoun, A. Michael Lincoff, Donna Ryan, et al. Semaglutide and cardiovascular risk reduction: A mediation analysis of the SELECT trial. *European Heart Journal*, page ehag524, 2026. doi: 10.1093/eurheartj/ehag524.

John Deanfield, A. Michael Lincoff, Steven E. Kahn, et al. Semaglutide and cardiovascular outcomes by baseline and changes in adiposity measurements: A prespecified analysis of the SELECT trial. *Lancet*, 406(10516):2257–2268, 2025. doi: 10.1016/S0140-6736(25)01375-3.

John Deanfield et al. Real-world, digital-first delivery of GLP-1 receptor agonist therapy in women. *Nature Medicine*, 2026. doi: 10.1038/s41591-026-04687-x.

Benjamin A. Goldstein, Nrupen A. Bhavsar, Matthew Phelan, and Michael J. Pencina. Controlling for informed presence bias due to the number of health encounters in an electronic health record. *American Journal of Epidemiology*, 184(11):847–855, 2016. doi: 10.1093/aje/kww112.

Deborah B. Horn et al. Cardiometabolic parameter change by weight regain on tirzepatide withdrawal in adults with obesity: A post hoc analysis of SURMOUNT-4. *JAMA Internal Medicine*, 186(2): 157–167, 2026. doi: 10.1001/jamainternmed.2025.6112.

Jennifer A. Hutcheon, Arnaud Chiolero, and James A. Hanley. Random measurement error and regression dilution bias. *BMJ*, 340:c2289, 2010. doi: 10.1136/bmj.c2289.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298

Harlan M. Krumholz, James A. de Lemos, Naveed Sattar, et al. Tirzepatide and blood pressure reduction: Stratified analyses of the SURMOUNT-1 randomised controlled trial. *Heart*, 110(19): 1165–1171, 2024. doi: 10.1136/heartjnl-2024-324170.

Hopin Lee et al. A guideline for reporting mediation analyses of randomized trials and observational studies: The AGReMA statement. *JAMA*, 326(11):1045–1056, 2021. doi: 10.1001/jama.2021.14075.

A. Michael Lincoff, Kirstine Brown-Frandsen, Helen M. Colhoun, et al. Semaglutide and cardiovascular outcomes in obesity without diabetes. *New England Journal of Medicine*, 389(24):2221–2232, 2023.

Marc Lipsitch, Eric Tchetgen Tchetgen, and Ted Cohen. Negative controls: A tool for detecting confounding and bias in observational studies. *Epidemiology*, 21(3):383–388, 2010. doi: 10.1097/EDE.0b013e3181d61eeb.

Yuan Lu et al. Changes in cardiovascular risk factors and health care expenditures among patients prescribed semaglutide. *JAMA Network Open*, 8(8):e2526013, 2025. doi: 10.1001/jamanetworkopen.2025.26013.

Judith E. Neter, Bianca E. Stam, Frans J. Kok, Diederick E. Grobbee, and Johanna M. Geleijnse. Influence of weight reduction on blood pressure: A meta-analysis of randomized controlled trials. *Hypertension*, 42(5):878–884, 2003. doi: 10.1161/01.HYP.0000094221.86888.AE.

Aleksandrina Ruseva et al. Semaglutide 2.4 mg long-term clinical outcomes in patients with obesity or overweight: Real-world evidence (SCOPE, 12 months). *Postgraduate Medicine*, 137(3–4): 251–260, 2025. doi: 10.1080/00325481.2025.2482274.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=7701298